

DejaView: Metric-Free Dynamic Spatial Memory for Mobile Robots in Changing Environments

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Abstract—Long-term robot autonomy requires spatial memory that remains searchable and updatable as objects move, disappear, and reappear. Dense metric semantic maps provide useful geometry, but maintaining globally consistent reconstruction and open-vocabulary features over large environments and long temporal horizons is expensive and susceptible to accumulated pose and depth errors. We present *DejaView*, a metric-free visual memory that consolidates RGB observations into semantic place cells and connects them as a topological graph. Each cell stores a place embedding, persistent object embeddings, and confidence values that support localized revision and time-aware forgetting. The resulting memory supports open-vocabulary object localization, contextual queries, graph search, and visual-waypoint navigation. Although the downstream navigation adapter uses depth and pose, memory construction and retrieval use RGB alone. Across static embodied benchmarks, *DejaView* matches geometry-dependent memory baselines. On *DejaZoo*, a real-world multi-traversal benchmark with controlled object changes, *DejaView* achieves 85.9% object localization recall, compared with 31.7% for the strongest evaluated metric semantic-map baseline. The graph also exhibits emergent spatial consistency. In on-robot navigation, sequential waypoint selection maintains 70% success through turns beyond 90°, where cyclic topological localization fails.

I. INTRODUCTION

Long-term deployment in human environments requires robots to remember not only where they have been, but also what was present and how the environment has changed. Objects are often relocated, and accurately perceiving these changes is complicated by viewpoint variation, partial occlusion, and perception noise. Consequently, a practical environment memory must support persistent recall, efficient search, and local revision without rebuilding the entire environment after each change.

Most robotic spatial memories are grounded in metric mapping. Classical SLAM provides geometry for localization and planning, while semantic mapping attaches object or language features to a reconstruction [1]–[3]. This design is powerful when pose and depth remain reliable, but errors can accumulate across long RGB streams and corrupt the semantic map. Recent object-centric and hierarchical scene graphs reduce semantic storage [4], [5], yet still generally assume posed RGB-D input or a reliable metric reconstruction. TSGM incrementally constructs a landmark-based topological semantic graph from RGB-D observations for scene representation and image-goal navigation [6]. RoboHop instead connects semantically meaningful image segments across observations to support open-vocabulary scene querying, planning, and navigation without metric poses [7].

DejaView explores a complementary design point: visual experience is organized directly in feature space as

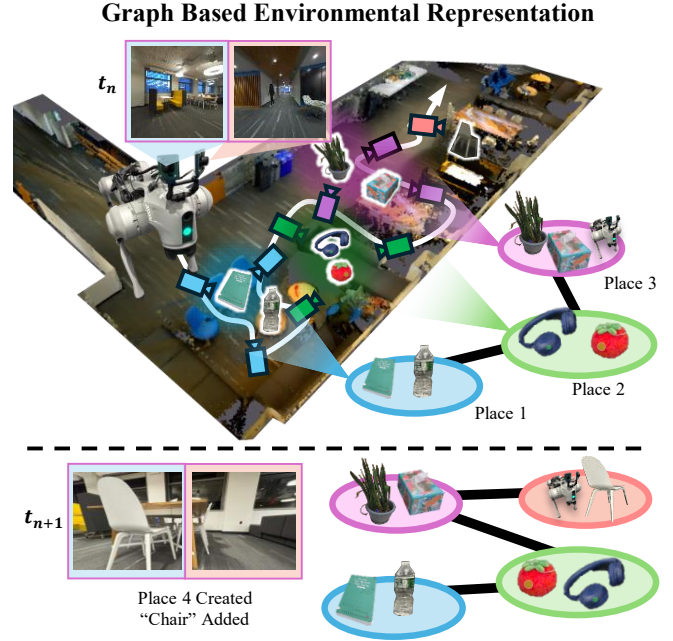


Fig. 1. **Graph-guided navigation.** The robot localizes itself and the target object, plans over the memory graph, and iteratively follows retrieved visual keyframes with a local navigation policy.

a compact graph of semantic place cells. Places serve as stable retrieval units, while object evidence is revised locally as new observations arrive. By removing the reliance on metric accuracy to assign object locations, *DejaView* aims to overcome compounding error and enable functional long horizon object tracking.

Our contributions are: 1) a metric-free, place-centric memory supporting persistent open-vocabulary recall, localized dynamic updates, and contextual queries from RGB; 2) *DejaZoo*, a multi-traversal benchmark with controlled object additions, removals, and relocations; and 3) real-world navigation experiments demonstrating that the memory graph can provide object localization, topological planning, and visual waypoints to different local controllers.

II. METRIC-FREE DYNAMIC MEMORY

A. Place Cells and Graph Construction

The atomic memory unit is a place cell $P_i = (\mathbf{e}_{p,i}, L_i)$, containing a keyframe place embedding $\mathbf{e}_{p,i}$ and a persistent object list L_i . Images from the robot’s fisheye cameras are encoded and concatenated into the current place embedding $\mathbf{e}_{p,t}$. While its cosine similarity to the active node remains above the place threshold ε_p , observations are associated with that node. When similarity falls below the threshold,

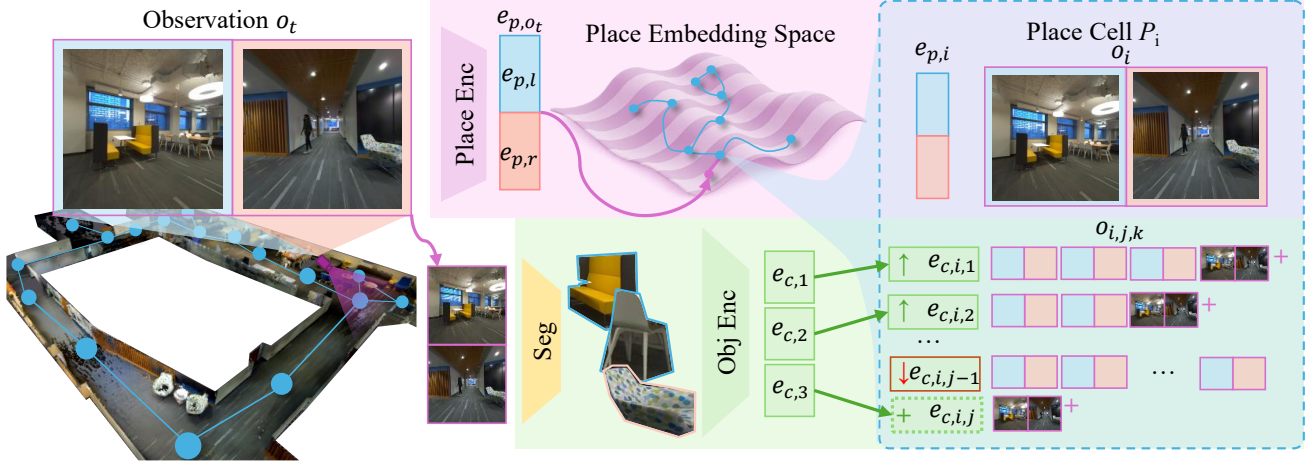


Fig. 2. **Memory construction.** At each timestep, *DejaView* compares the current place embedding with stored nodes for continued localization, loop closure, or new-node creation. Segmented object observations are consolidated into the active place cell, while unobserved entries lose confidence.

DejaView searches for a loop closure among stored nodes; otherwise, it creates a new place cell and connects it to the previous node. Edges encode topological adjacency rather than metric distance.

B. Object Consolidation and Dynamics

Segmented object crops are encoded into visual-semantic embeddings and buffered during a visit. On departure, a buffered embedding $\mathbf{e}_b$ is merged with its nearest object entry $\mathbf{e}_l$ when their similarity exceeds the object threshold ε_o :

$$\mathbf{e}_l \leftarrow \frac{n_l \mathbf{e}_l + \mathbf{e}_b}{n_l + 1}, \quad \mathbf{e}_b^\top \mathbf{e}_l > \varepsilon_o, \quad (1)$$

where n_l is the number of prior observations. Otherwise, a new object entry is appended. Each object carries confidence $c_o \in [0, 1]$. Re-observation refreshes c_o , while a missed observation decays it as $c_o \leftarrow \gamma c_o$. This prevents a single occlusion or detection failure from immediately deleting persistent memory.

Severe visual changes can cause a new place cell to be created. The old cell is retained but becomes less likely to be revisited, providing an implicit form of forgetting without requiring explicit geometric alignment between old and new reconstructions.

C. Querying and Navigation

For an object query embedding $\mathbf{e}_q$, candidates are ranked by semantic similarity, observation confidence, and recency:

$$s_q(\mathbf{e}_c) = (\mathbf{e}_c^\top \mathbf{e}_q) c_o(\mathbf{e}_c) \exp(-\Delta t / \tau). \quad (2)$$

Contextual queries additionally rerank candidates using the similarity between the full language query and other objects co-located in the same place cell. This introduces useful place-level context without relation supervision.

For navigation, the robot localizes itself and the queried object to graph nodes, searches for a path, and passes the corresponding keyframes to a local image-goal controller. Edge costs use adjacent place-embedding distance, $d_{ij} = 1 - \mathbf{e}_{p,i}^\top \mathbf{e}_{p,j}$.

III. EXPERIMENTS

A. Static Embodied QA Benchmark

On GOAT-Bench Val Unseen, we evaluate *DejaView* on 278 subtasks while retaining memory across sequential goals. Its semantic memory is constructed and queried from RGB alone; depth and pose are used only by downstream TSDF navigation. *DejaView* achieves 69.06% Success and 49.52% SPL, matching 3D-Mem’s reported 69.1% Success and exceeding its 48.9% SPL despite using no geometry in memory.

TABLE I
GOAT-BENCH VAL UNSEEN. †: PAPER SUBSET; ‡: OUR 278-SUBTASK EVALUATION. *DejaView* MEMORY IS RGB-ONLY; DOWNSTREAM NAVIGATION USES DEPTH AND POSE.

Method	Memory Input	Success Rate $\uparrow$	SPL $\uparrow$
<i>Open-Sourced VLM Exploration</i>			
3D-Mem w/o memory†	RGB-D + pose	40.6	14.6
3D-Mem†	RGB-D + pose	49.6	29.4
<i>GPT-4o Exploration</i>			
Explore-EQA†	RGB-D + pose	55.0	37.9
CG w/ Frontier Snapshots†	RGB-D + pose	61.5	45.3
3D-Mem w/o memory†	RGB-D + pose	58.6	38.5
3D-Mem†	RGB-D + pose	69.1	48.9
DejaView (Ours)‡	RGB only	69.06	49.52

B. Dynamic-Memory Benchmark

DejaZoo contains three floor-scale indoor scenes, each exceeding 100 m², with two or three traversals per environment. Between traversals, objects are added, removed, or relocated. The benchmark tracks more than 30 unique objects and contains over 15,000 human object annotations. High-fidelity scanner reconstructions provide geometric ground truth for analysis, but *DejaView* itself receives only RGB observations.

C. Object Localization

At the end of each traversal, the system is queried for each annotated object. A retrieval is correct when the returned place or voxel corresponds to the object’s most recent annotated location. Table II compares *DejaView* with metric semantic memories and reconstruction-based pipelines.

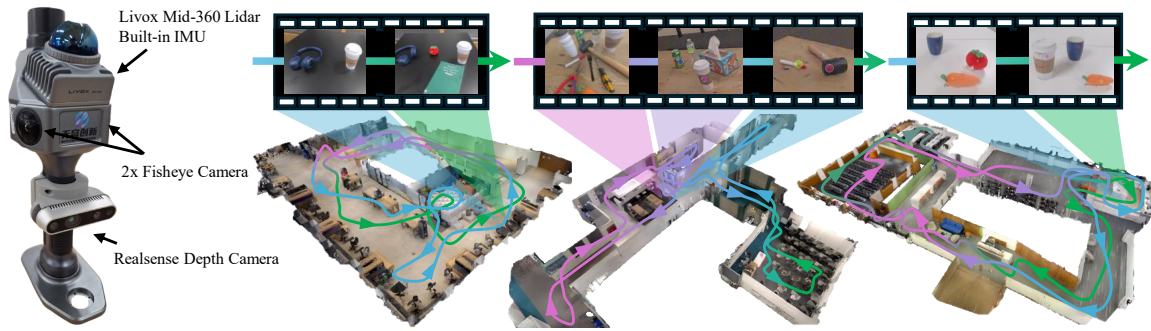


Fig. 3. *DejaZoo* dynamic environments. Repeated traversals cover three large indoor scenes with controlled object additions, removals, and relocations. Scanner reconstructions are used for evaluation only.

TABLE II

OBJECT LOCALIZATION RECALL (%). FULL SEQUENCES AND THE SMALLER-SCENE DIAGNOSTIC.

Method	Input	Full	Small
DynaMem [8]	RGB-D + pose	27.6	74.1
VGGT-SLAM [9]	RGB	11.8	51.4
DepthAnything3 [10]	RGB	10.4	59.8
VGGT-SLAM2 [11]	RGB	28.9	44.3
HOV-SG [5]	RGB-D + pose	27.3	54.0
ConceptGraphs [4]	RGB-D + pose	31.7	58.8
<i>DejaView</i> (ours)	RGB	85.9	84.2

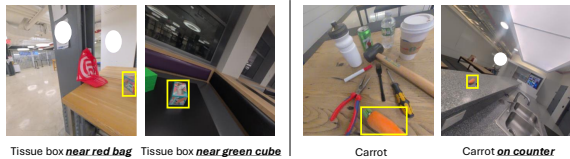


Fig. 4. **Contextual object retrieval.** Place-cell co-occurrence changes the selected observation for queries such as “tissue box near red bag” and “carrot on counter” without explicit relation labels.

DejaView reaches 85.9% recall on full *DejaZoo*, while the strongest baseline, ConceptGraphs, reaches 31.7%.

Metric baselines improve substantially on smaller scenes, indicating that their semantic representations are useful when pose and reconstruction remain reliable. Their degradation on the full benchmark is dominated by maintaining consistent geometry across long, changing sequences. *DejaView* remains stable because semantic memory is not tied to a globally reconstructed coordinate frame.

D. Context and Threshold Robustness

Fig. 4 shows that co-located objects provide enough context to distinguish otherwise similar category queries. The method remains limited by place-cell granularity: fine relations between visually similar objects in the same cell may remain ambiguous.

Fig. 5 shows that *DejaView* is robust across a broad range of place and object similarity thresholds. In the practical cosine-similarity range of 0.6–0.9, object recall and place localization remain strong, while memory growth and update cost increase gradually rather than causing abrupt performance degradation. We use $\varepsilon_p = 0.7$ and $\varepsilon_o = 0.8$ for all

TABLE III

MEMORY FOR IMAGE GOAL NAVIGATION: OFFLINE REPLAY VS. ON-ROBOT GOAL SELECTION.

Method	Offline MAE ↓	$\approx 45^\circ$ Turns	$> 90^\circ$ Turn
Topological [12]	0.93	6/10	0/10
Ours (Sequential)	0.55	7/10	7/10

experiments; nearby settings produce comparable retrieval quality.

E. Emergent Spatial Consistency

The memory graph contains no metric coordinates, but Metacam ground truth allows us to test whether its topology corresponds to physical structure. Maximum within-cell distances concentrate near zero, with larger cells corresponding to large feature-sparse environments (such as a stretch of hallway). Edge distances span a broader range, showing emergent spatial separation rather than arbitrary visual grouping.

F. Goal Selection Strategies

DejaView can serve as topological scene memory for navigation policies. We compare our sequential thresholding with NoMaD’s cyclic topological localization [12], both using NoMaD policy as the navigation backbone for action generation. Our method advances a monotonic route index when predicted goal distance crosses a threshold, with one-node lookahead; the baseline relocalizes within a ± 2 -node window and may move backward.

Table III exposes a gap between replay and deployment. Sequential thresholding lowers offline node error (0.55 vs. 0.93) and maintains 70% success on both moderate and sharp turns, whereas topological localization drops from 60% to 0% beyond 90° . Replay advances observations regardless of commanded actions, masking control failures during sharp maneuvers.

IV. LIMITATIONS AND DISCUSSION

DejaView depends on off-the-shelf segmentation, visual-language embeddings, and visual place recognition. Perception failures can duplicate one object across cells, merge visually similar instances, or create false loop closures. Confidence decay handles uncertain absence but is not a calibrated probabilistic model of object state. Moreover, a topological graph does not replace metric geometry when

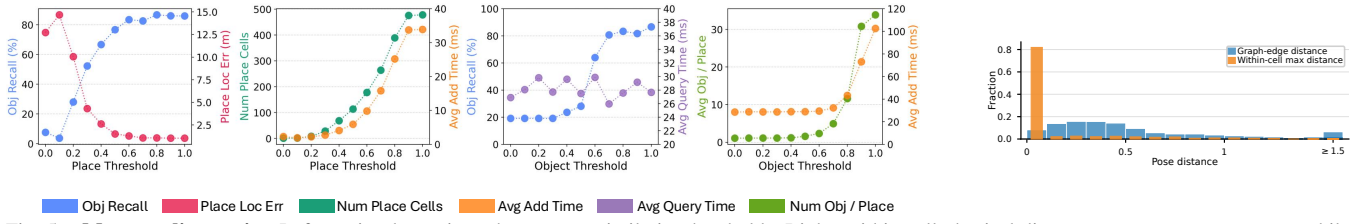


Fig. 5. **Memory diagnostics.** Left: retrieval remains robust across similarity thresholds. Right: within-cell physical distances concentrate near zero, while graph edges span larger distances. Geometry is used only for evaluation.

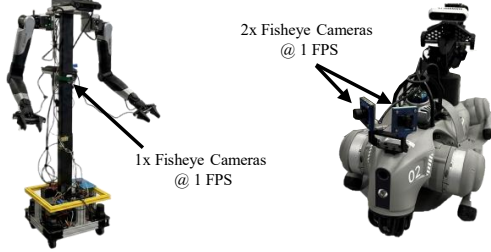


Fig. 6. **Evaluation robots.** *DejaView* runs on YOR (left), with one front-facing fisheye camera, and Unitree Go2 (right), with two head-mounted fisheye cameras.

V. CONCLUSION

DejaView organizes RGB experience into a compact, dynamic graph of semantic place cells. It supports localized memory updates, open-vocabulary and contextual retrieval, and graph-guided navigation without requiring a dense metric semantic map. Strong recall on changing, multi-traversal environments and robust sequential waypoint selection in on-robot trials indicate that metric-free memory is a practical complement to geometric mapping for long-term robot autonomy.

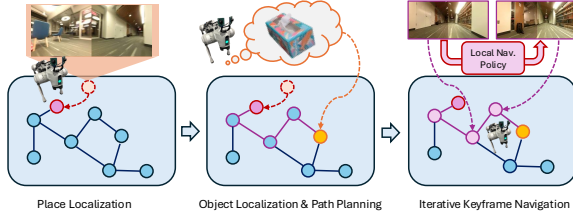


Fig. 7. **Visual-waypoint navigation.** Place localization, graph planning, and iterative local control.

collision avoidance, precise manipulation, or fine spatial relations are required.

These limitations suggest hybrid extensions rather than a categorical rejection of metric mapping. Metric estimates can be attached selectively to place cells, object association can incorporate explicit temporal uncertainty, and memory retrieval can be coupled more tightly with learned local control. The present results establish that feature-space memory is a strong foundation for long-horizon semantic recall in regimes where dense reconstruction is difficult.

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